# MA 252 Final Project

## Analysis of College Major Salaries

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**Date:** May 4, 2026

**Contributions:**

- **Kayla:** Data cleaning, exploratory data analysis, initial model setup
- **Elizabeth:** Regression modeling, visualizations, interpretation, written analysis

## Abstract:

This project analyzes the factors that influence median salary outcomes for recent college graduates using the FiveThirtyEight "recent-grads" dataset. The analysis focuses on how gender composition, unemployment rate, and major category relate to earnings across 172 academic majors. A simple linear regression was first used to examine the relationship between median salary and the proportion of women in a major. A multiple regression model was then fit to add in the unemployment rate and major category, followed by stepwise model selection to identify the most effective combination of predictors. The results show that major category is the strongest predictor of salary, while the proportion of women remains a significant negative predictor even after adjusting for field of study. Unemployment rate does not meaningfully contribute once other variables are included. Overall, the final model explains over 62% of the variation in median salary, showing substantial differences in earnings across academic fields.


## Introduction:

Choosing a college major is one of the most important decisions students make, and it can have long-lasting effects on employment opportunities and salaries. Understanding how different characteristics of academic fields relate to salary outcomes can help students, educators, and others make more educated decisions. This project uses the FiveThirtyEight "recent-grads" dataset, which contains information on 172 majors, including median salary, gender composition, unemployment rate, and major category. The goal of this analysis is to investigate which factors best explain variation in median salary across majors. Specifically, the study examines whether the proportion of women in a major, the unemployment rate, and the academic field (major category) are significant predictors of earnings. To manage these questions, the analysis starts with a simple linear regression model, followed by a multiple regression model that adds additional predictors. A stepwise model selection procedure is also used to identify the most effective combination of variables. By combining our results with what we learned from the diagnostics, this project gives a clearer picture of how major characteristics relate to salary outcomes. The data highlights significant differences across academic fields and reveals how gender composition, differences across majors, and other external factors help explain earnings differences.

```
#uploading the data
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.2
## v ggplot2   4.0.2     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.1.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)
library(car)
```

```
## Loading required package: carData
##
## Attaching package: 'car'
##
## The following object is masked from 'package:dplyr':
```

```
##
##     recode
##
## The following object is masked from 'package:purrr':
##
##     some
```

```
library(skimr)
library(moderndive)
```

```
url <- "https://raw.githubusercontent.com/fivethirtyeight/data/master/college-majors/recent-grads.csv"

college_major_data <- read_csv(url)
```

```
## Rows: 173 Columns: 21
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (2): Major, Major_category
## dbl (19): Rank, Major_code, Total, Men, Women, ShareWomen, Sample_size, Empl...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
college_major_data <- na.omit(college_major_data)

view(college_major_data)
```

```
# Dataset structure
head(college_major_data)
```

```
## # A tibble: 6 x 21
##    Rank Major_code Major Total   Men Women Major_category ShareWomen Sample_size
##   <dbl>      <dbl> <chr> <dbl> <dbl> <dbl> <chr>               <dbl>       <dbl>
## 1     1       2419 PETR~  2339  2057   282 Engineering         0.121          36
## 2     2       2416 MINI~   756   679    77 Engineering         0.102           7
## 3     3       2415 META~   856   725   131 Engineering         0.153           3
## 4     4       2417 NAVA~  1258  1123   135 Engineering         0.107          16
## 5     5       2405 CHEM~ 32260 21239 11021 Engineering         0.342         289
## 6     6       2418 NUCL~  2573  2200   373 Engineering         0.145          17
## # i 12 more variables: Employed <dbl>, Full_time <dbl>, Part_time <dbl>,
## #   Full_time_year_round <dbl>, Unemployed <dbl>, Unemployment_rate <dbl>,
## #   Median <dbl>, P25th <dbl>, P75th <dbl>, College_jobs <dbl>,
## #   Non_college_jobs <dbl>, Low_wage_jobs <dbl>
```

```
glimpse(college_major_data)
```

```
## Rows: 172
## Columns: 21
## $ Rank                <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15~
## $ Major_code          <dbl> 2419, 2416, 2415, 2417, 2405, 2418, 6202, 5001, 2~
## $ Major               <chr> "PETROLEUM ENGINEERING", "MINING AND MINERAL ENGI~
```

```
## $ Total                <dbl> 2339, 756, 856, 1258, 32260, 2573, 3777, 1792, 91~
## $ Men                  <dbl> 2057, 679, 725, 1123, 21239, 2200, 2110, 832, 803~
## $ Women                <dbl> 282, 77, 131, 135, 11021, 373, 1667, 960, 10907, ~
## $ Major_category       <chr> "Engineering", "Engineering", "Engineering", "Eng~
## $ ShareWomen           <dbl> 0.12056434, 0.10185185, 0.15303738, 0.10731320, 0~
## $ Sample_size          <dbl> 36, 7, 3, 16, 289, 17, 51, 10, 1029, 631, 399, 14~
## $ Employed             <dbl> 1976, 640, 648, 758, 25694, 1857, 2912, 1526, 764~
## $ Full_time            <dbl> 1849, 556, 558, 1069, 23170, 2038, 2924, 1085, 71~
## $ Part_time            <dbl> 270, 170, 133, 150, 5180, 264, 296, 553, 13101, 1~
## $ Full_time_year_round <dbl> 1207, 388, 340, 692, 16697, 1449, 2482, 827, 5463~
## $ Unemployed           <dbl> 37, 85, 16, 40, 1672, 400, 308, 33, 4650, 3895, 2~
## $ Unemployment_rate    <dbl> 0.018380527, 0.117241379, 0.024096386, 0.05012531~
## $ Median               <dbl> 110000, 75000, 73000, 70000, 65000, 65000, 62000,~
## $ P25th                <dbl> 95000, 55000, 50000, 43000, 50000, 50000, 53000, ~
## $ P75th                <dbl> 125000, 90000, 105000, 80000, 75000, 102000, 7200~
## $ College_jobs         <dbl> 1534, 350, 456, 529, 18314, 1142, 1768, 972, 5284~
## $ Non_college_jobs     <dbl> 364, 257, 176, 102, 4440, 657, 314, 500, 16384, 1~
## $ Low_wage_jobs        <dbl> 193, 50, 0, 0, 972, 244, 259, 220, 3253, 3170, 98~
```

```
nrow(college_major_data)
```

```
## [1] 172
```

## Data Characteristics:

We used FiveThirtyEight's recent college graduates dataset, which includes 173 majors and their employment and salary outcomes. After importing the data from GitHub, we removed rows with missing values so the analysis was based on a complete dataset. The variables used in this project include median salary (response), gender composition, unemployment rate, and major category.

```
#visualizing the data
college_major_data %>% select(Median) %>% skim()
```

Table 1: Data summary

| | |
|---|---|
| Name | Piped data |
| Number of rows | 172 |
| Number of columns | 1 |
| Column type frequency: | |
| numeric | 1 |
| Group variables | None |

**Variable type: numeric**

| skim_variable | n_missing | complete_rate | mean | sd | p0 | p25 | p50 | p75 | p100 | hist |
|---|---|---|---|---|---|---|---|---|---|---|
| Median | 0 | 1 | 40076.74 | 11461.39 | 22000 | 33000 | 36000 | 45000 | 110000 | |

```r
# Summary statistics for predictors
college_major_data %>%
  select(ShareWomen, Unemployment_rate, Total) %>%
  skim()
```

Table 3: Data summary

| | |
|---|---|
| Name | Piped data |
| Number of rows | 172 |
| Number of columns | 3 |
| Column type frequency: | |
| numeric | 3 |
| Group variables | None |

**Variable type: numeric**

| skim_variable | n_missing | complete_rate | mean | sd | p0 | p25 | p50 | p75 | p100 | hist |
|---|---|---|---|---|---|---|---|---|---|---|
| ShareWomen | 0 | 1 | 0.52 | 0.23 | 0 | 0.34 | 0.53 | 0.70 | 0.97 | |
| Unemployment_rate | 0 | 1 | 0.07 | 0.03 | 0 | 0.05 | 0.07 | 0.09 | 0.18 | |
| Total | 0 | 1 | 39370.08 | 63483.49 | 124 | 4549.75 | 15104.00 | 38909.75 | 393735.00 | |

## Exploratory Data Analysis:

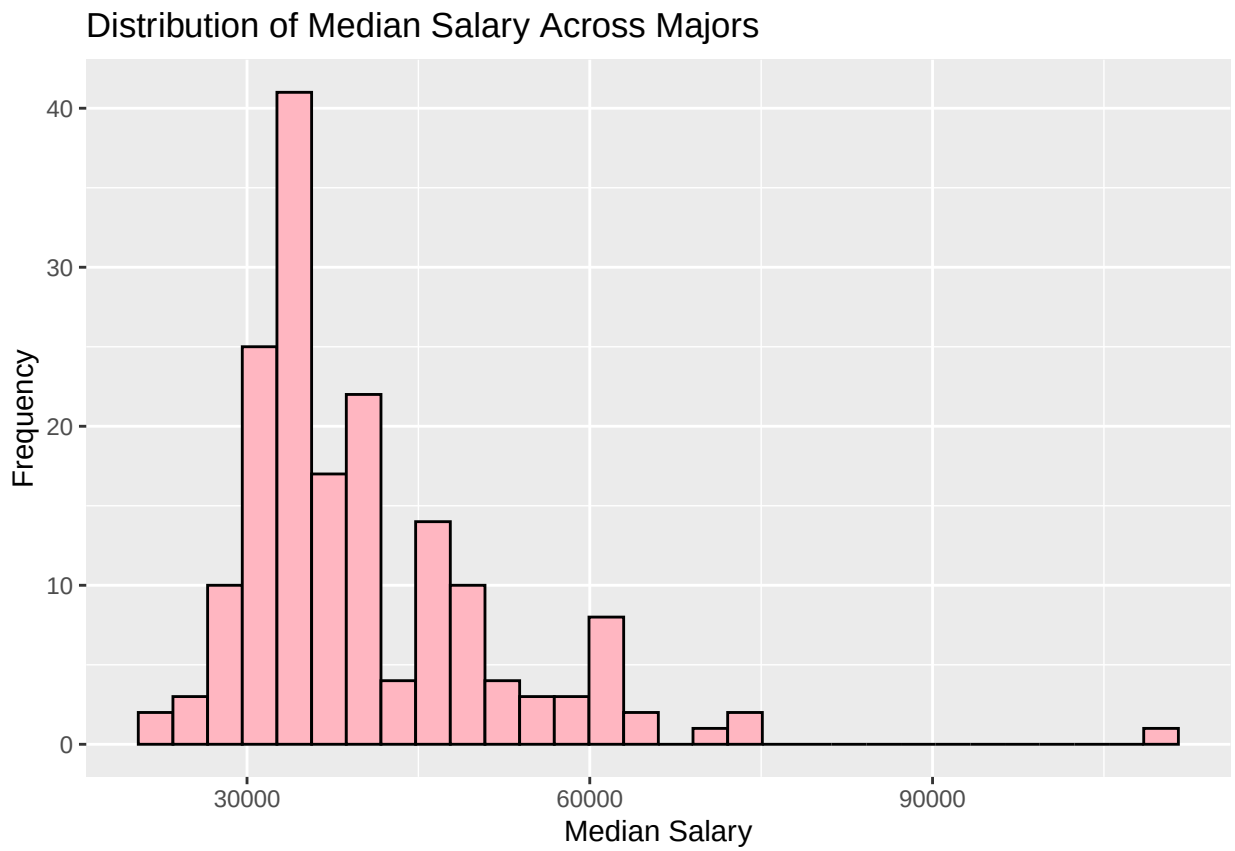
To understand the key variables in our dataset, we examined the distribution of the response variable (median salary) and three predictors: ShareWomen, Unemployment_rate, and Total. The output helped us understand the overall shape, spread, and center of each variable.

For median salary, the mean is about $40,076, while the median is $36,000, indicating a slight right skew. Salaries range from $22,000 to $110,000, so there's a lot of variation across majors. The interquartile range (from $33,000 to $45,000) shows that half of all majors fall within that salary band.

For the predictors, ShareWomen has a mean of 0.52, meaning women make up about half of graduates in the average major. The values range from 0.00 to 0.97, showing big differences in gender representation across fields. Unemployment_rate is somewhat low overall, with a mean of 0.07 and most majors falling between 0.05 and 0.09. Total, which measures the number of graduates in each major, ranges from 124 to 393,735, showing that some majors are extremely popular while others are much less so.

Overall, these summaries show that the dataset exhibits substantial variation in both salary outcomes and the predictors, making it good for regression analysis.

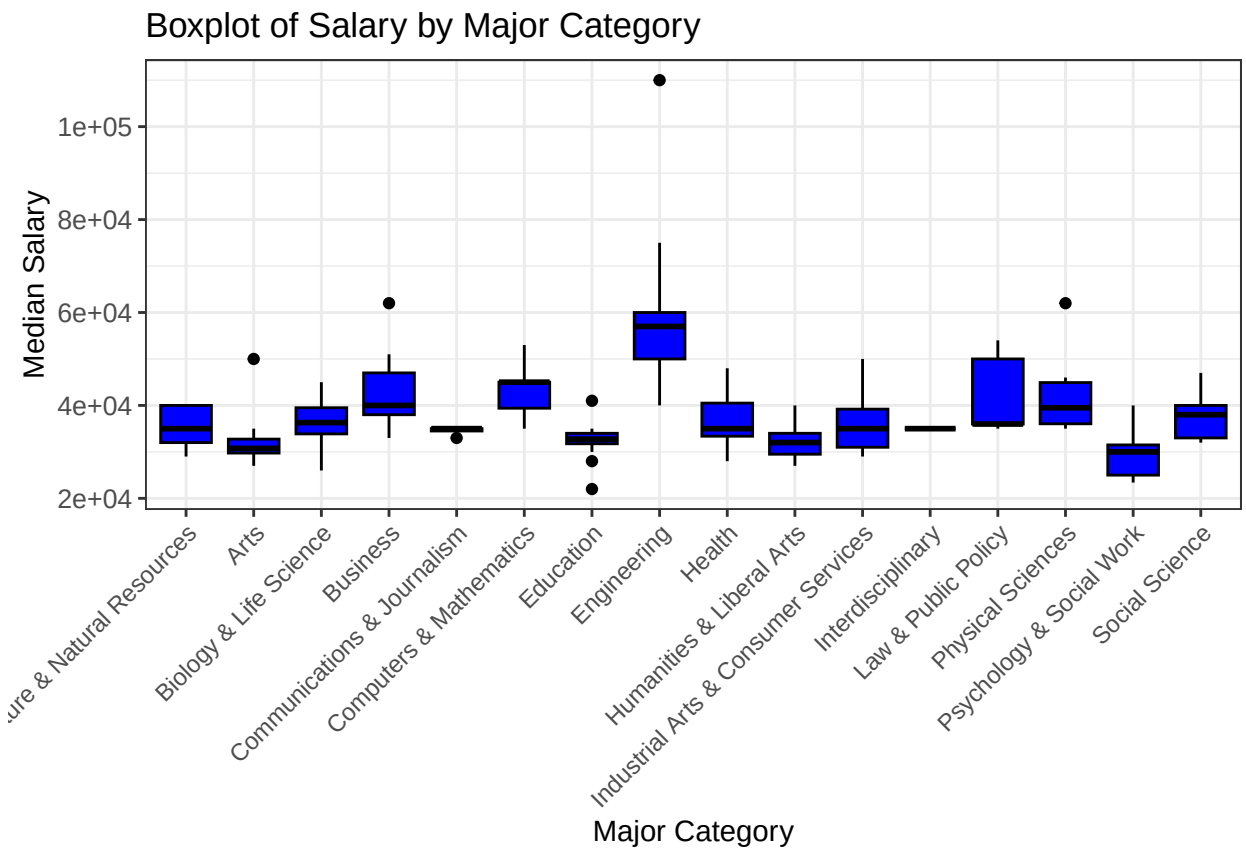
```r
#histogram of response variable
ggplot(college_major_data, aes(x=Median)) +
  geom_histogram(fill="lightpink", color="black", bins = 30) +
  labs(title = "Distribution of Median Salary Across Majors", x="Median Salary", y="Frequency")
```




**Distribution of Median Salary:**

The histogram shows how median salaries are spread across majors. The distribution is right-skewed, most majors fall between $30,000 and $50,000, and only a smaller group earns much higher salaries. This matches what we saw in the summary stats since the median salary is $36,000, but the maximum goes up to $110,000. That big gap shows that a few high-earning majors pull the mean upward, even though most majors sit in a more moderate salary range.

```
ggplot(college_major_data, aes(x=Major_category, y=Median)) +
  geom_boxplot(fill = "blue", color = "black") +
  labs(title = "Boxplot of Salary by Major Category",
       x = "Major Category",
       y = "Median Salary"
 ) +
  theme_bw() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1, size = 9),
    axis.text.y = element_text(size = 10)
  )
```




**Salary Differences Across Major Categories:**

To compare salary outcomes across different fields of study, we made a boxplot of median salary by major category. The plot shows that earnings vary a lot across majors. In areas like Engineering, Computers & Mathematics, and Business, salaries tend to be higher on the scale, while fields such as Education, Psychology & Social Work, and Humanities & Liberal Arts fall on the lower end with less variability.

The amount of spread within each category also differs. Engineering, for example, has a wider interquartile range, meaning salaries vary quite a bit across different engineering majors. However, fields like Education have a much narrower range, showing that earnings are more consistent within that category.

Overall, this graph makes it clear that salary outcomes differ heavily across academic fields, supporting the idea that major choice plays a big role in post-grad salaries.

```
simple_model <- lm(Median ~ ShareWomen, data = college_major_data)
summary(simple_model)
```

```
##
## Call:
## lm(formula = Median ~ ShareWomen, data = college_major_data)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -17261  -5474  -1007   3502  57604
##
## Coefficients:
```

```
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)    56093       1705   32.90   <2e-16 ***
## ShareWomen    -30670       2987  -10.27   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9031 on 170 degrees of freedom
## Multiple R-squared:  0.3828, Adjusted R-squared:  0.3791
## F-statistic: 105.4 on 1 and 170 DF,  p-value: < 2.2e-16
```

$$\widehat{Median} = \beta_0 + \beta_1(\text{ShareWomen})$$

**Simple Linear Regression Model**

The slope in this model is –30,670. This means that majors with a higher proportion of women tend to have lower predicted median salaries in this dataset, and majors with fewer women tend to have higher predicted salaries.

The intercept of $56,093 represents the predicted salary for a major with 0% women. ShareWomen has a very small p-value (< 2e-16), so the relationship is statistically significant.

The model's $R^2$ is 0.3828, meaning ShareWomen alone explains about 38% of the variation in median salary across majors. It picks up a meaningful part of the variation, but it's not enough on its own to explain all the differences in salaries across majors.

```r
multiple_model <- lm(Median ~ ShareWomen + Unemployment_rate + factor(Major_category),
                     data = college_major_data)
summary(multiple_model)
```

```
##
## Call:
## lm(formula = Median ~ ShareWomen + Unemployment_rate + factor(Major_category),
##     data = college_major_data)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -20053  -3866   -739   2623  49462
##
## Coefficients:
##                                                   Estimate Std. Error
## (Intercept)                                        42845.6     3142.5
## ShareWomen                                        -15237.7     4207.5
## Unemployment_rate                                 -30090.5    21318.4
## factor(Major_category)Arts                          2128.5     3774.9
## factor(Major_category)Biology & Life Science        4356.3     3265.2
## factor(Major_category)Business                     10194.0     3258.8
## factor(Major_category)Communications & Journalism   3959.6     4604.1
## factor(Major_category)Computers & Mathematics       7185.8     3439.5
## factor(Major_category)Education                     2465.6     3414.9
## factor(Major_category)Engineering                  20083.0     2935.2
## factor(Major_category)Health                        8079.2     3661.4
## factor(Major_category)Humanities & Liberal Arts     1132.3     3313.7
## factor(Major_category)Industrial Arts & Consumer Services   269.6     3750.3
```

```
## factor(Major_category)Interdisciplinary                      6033.3     7980.4
## factor(Major_category)Law & Public Policy                    9456.4     4231.2
## factor(Major_category)Physical Sciences                      8195.1     3443.2
## factor(Major_category)Psychology & Social Work               1527.6     3870.5
## factor(Major_category)Social Science                         5820.4     3661.4
##                                                            t value Pr(>|t|)
## (Intercept)                                                 13.634  < 2e-16 ***
## ShareWomen                                                  -3.622 0.000397 ***
## Unemployment_rate                                           -1.411 0.160120
## factor(Major_category)Arts                                   0.564 0.573665
## factor(Major_category)Biology & Life Science                 1.334 0.184117
## factor(Major_category)Business                               3.128 0.002104 **
## factor(Major_category)Communications & Journalism            0.860 0.391121
## factor(Major_category)Computers & Mathematics                2.089 0.038334 *
## factor(Major_category)Education                              0.722 0.471384
## factor(Major_category)Engineering                            6.842 1.73e-10 ***
## factor(Major_category)Health                                 2.207 0.028825 *
## factor(Major_category)Humanities & Liberal Arts              0.342 0.733052
## factor(Major_category)Industrial Arts & Consumer Services    0.072 0.942779
## factor(Major_category)Interdisciplinary                      0.756 0.450794
## factor(Major_category)Law & Public Policy                    2.235 0.026862 *
## factor(Major_category)Physical Sciences                      2.380 0.018533 *
## factor(Major_category)Psychology & Social Work               0.395 0.693628
## factor(Major_category)Social Science                         1.590 0.113959
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7427 on 154 degrees of freedom
## Multiple R-squared:  0.6219, Adjusted R-squared:  0.5801
## F-statistic:  14.9 on 17 and 154 DF,  p-value: < 2.2e-16
```

$$\widehat{Median} = \beta_0 + \beta_1(\text{ShareWomen}) + \beta_2(\text{UnemploymentRate}) + \sum_{k=1}^{K-1} \beta_{3k}(\text{MajorCategory}_k)$$

$$\text{MajorCategory}_{ik} = \begin{cases} 1, & \text{if major } i \text{ is in category } k \\ 0, & \text{otherwise} \end{cases}$$

**Multiple Linear Regression Model:**

To improve the simple model, we fit a multiple linear regression that included ShareWomen, Unemployment_rate, and Major_category as predictors of median salary. Adding these variables helps the model account for differences across fields of study and market conditions.

After adjusting for the other predictors, the coefficient for ShareWomen becomes –15,237.7, which is smaller than in the simple model. This suggests that part of the negative relationship between gender and salary is explained by differences across major categories. However, the effect is still statistically significant ($p = 0.000397$), meaning majors with a higher proportion of women still tend to have lower median salaries after controlling for the other variables.

The coefficient for Unemployment_rate is –30,090.5, but it isn't statistically significant ($p = 0.160$). This shows that once we account for major category and gender, unemployment rate doesn't meaningfully predict median salary across majors.

Major_category shows clear differences in earnings across fields. Several categories such as Engineering, Business, Computers & Mathematics, Health, Law & Public Policy, and Physical Sciences, have

statistically significant positive coefficients, meaning these majors tend to earn higher salaries than the original category. Engineering stands out with a large and highly significant effect, consistant with what we saw in the earlier boxplots.

The model's $R^2$ is 0.6219, meaning that about 62.2% of the variation in median salary is explained by the combination of gender composition, unemployment rate, and major category. This is a significant improvement over the simple model's $R^2$ of 0.3828, showing that including the major category greatly strengthens the model.

Overall, the multiple regression results show that field of study is the strongest predictor of salary, and gender composition remains significant even after adjusting for major category. Unemployment rate doesn't appear to play an important role once the other factors are included.

```
#Stepwise Model Selection
model_empty <- lm(Median ~ 1, data = college_major_data)
model_stepwise <- step(model_empty, direction = "both",
                       scope = ~ ShareWomen + Unemployment_rate + Major_category + Total)
```

```
## Start:  AIC=3216.28
## Median ~ 1
##
##                       Df  Sum of Sq        RSS    AIC
## + Major_category    15 1.3070e+10 9.3931e+09 3096.3
## + ShareWomen         1 8.5984e+09 1.3865e+10 3135.3
## + Unemployment_rate  1 3.0106e+08 2.2162e+10 3216.0
## <none>                             2.2463e+10 3216.3
## + Total              1 2.5592e+08 2.2207e+10 3216.3
##
## Step:  AIC=3096.31
## Median ~ Major_category
##
##                       Df  Sum of Sq        RSS    AIC
## + ShareWomen         1 7.8951e+08 8.6036e+09 3083.2
## + Unemployment_rate  1 1.7603e+08 9.2170e+09 3095.1
## <none>                             9.3931e+09 3096.3
## + Total              1 1.5826e+07 9.3772e+09 3098.0
## - Major_category    15 1.3070e+10 2.2463e+10 3216.3
##
## Step:  AIC=3083.21
## Median ~ Major_category + ShareWomen
##
##                       Df  Sum of Sq        RSS    AIC
## + Unemployment_rate  1  109881631 8.4937e+09 3083.0
## <none>                             8.6036e+09 3083.2
## + Total              1    4624260 8.5989e+09 3085.1
## - ShareWomen         1  789507935 9.3931e+09 3096.3
## - Major_category    15 5261219886 1.3865e+10 3135.3
##
## Step:  AIC=3083
## Median ~ Major_category + ShareWomen + Unemployment_rate
##
##                       Df  Sum of Sq        RSS    AIC
## <none>                             8.4937e+09 3083.0
## - Unemployment_rate  1  109881631 8.6036e+09 3083.2
## + Total              1    4359397 8.4893e+09 3084.9
```

```
## - ShareWomen          1  723361833 9.2170e+09 3095.1
## - Major_category     15 5258922169 1.3753e+10 3135.9
```

## Stepwise Model Selection:

To see which predictors were most useful for explaining median salary, we used stepwise model selection. The procedure tested the available variables ShareWomen, Unemployment_rate, Major_category, and Total and chose the ones that improved the model based on AIC.

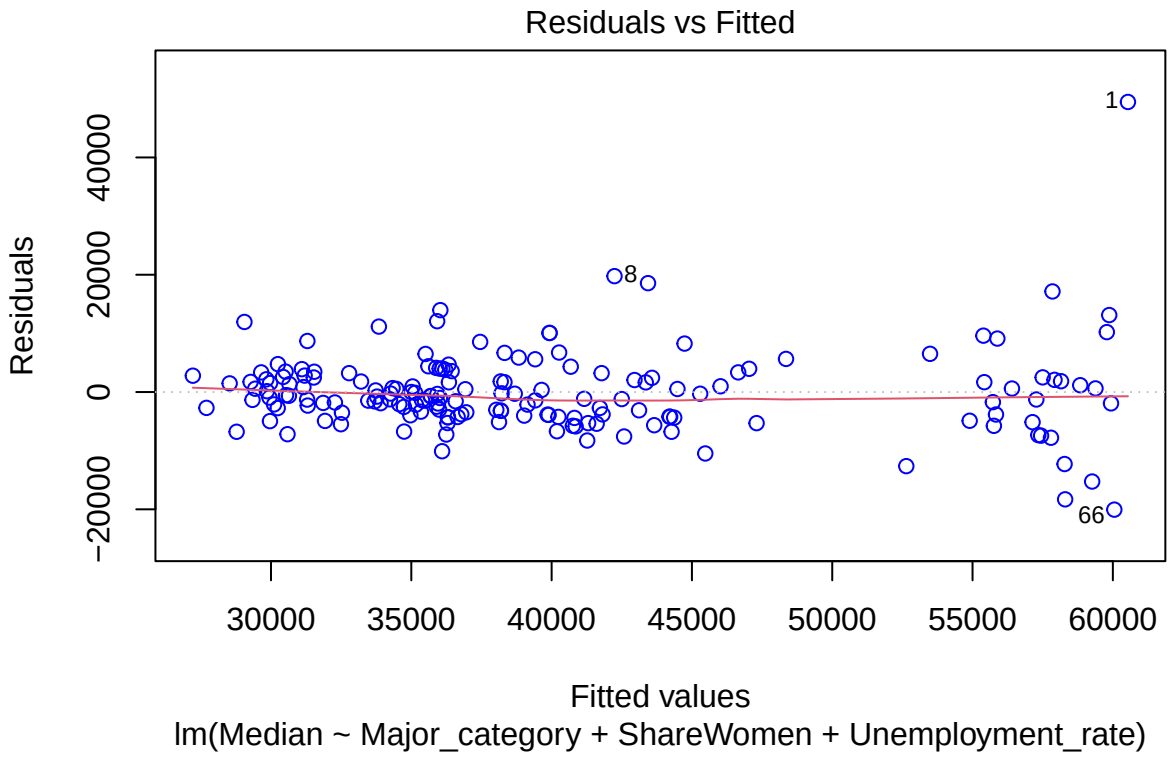
The first variable selected was Major_category, which made the biggest improvement in AIC. This confirms that field of study is the strongest predictor of salary. After that, we added ShareWomen, which also improved the model. Unemployment_rate only gave a small improvement, but it was still included in the final model. Total, which measures the number of graduates in each major, didn't help the model and wasn't selected.

The final stepwise model includes:

- Major_category
- ShareWomen
- Unemployment_rate

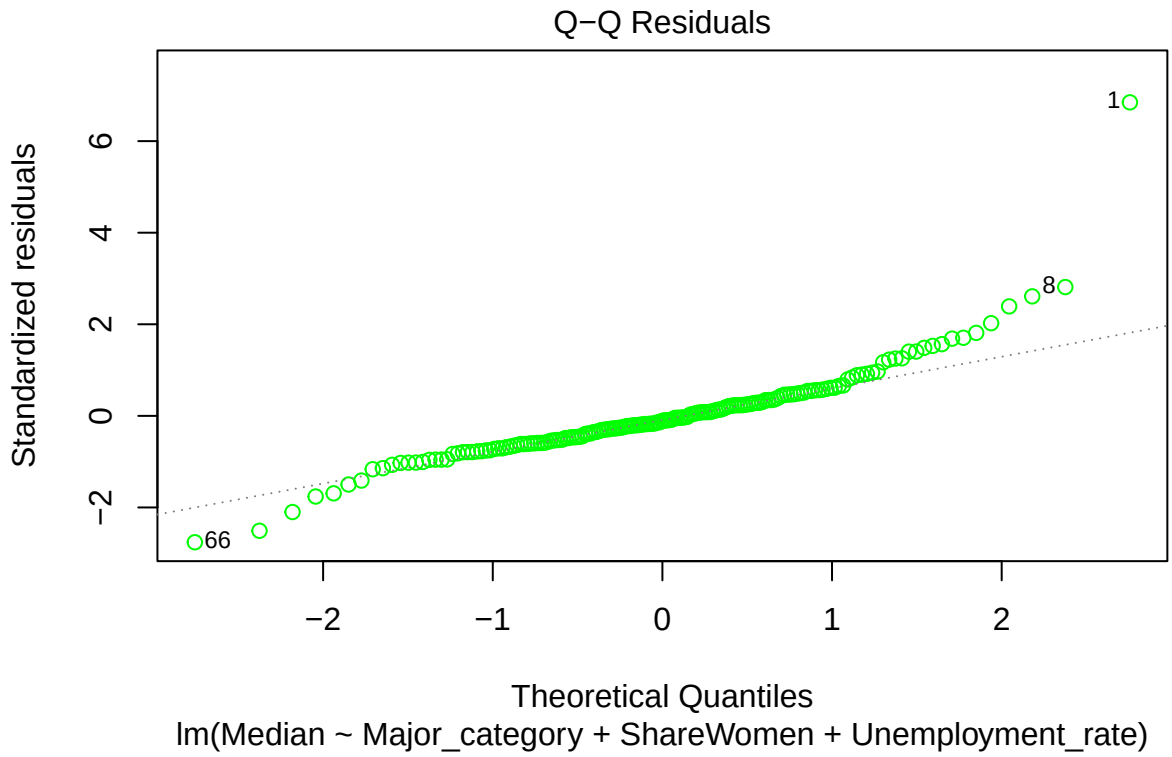
This model is very similar to the multiple regression model we fit earlier, and it has a much lower AIC than the simple model. The stepwise results reinforce that major category is the most important predictor, and ShareWomen still matters even after adjusting for field of study. Total doesn't contribute much to predicting median salary.

```
#Model Assumptions
plot(model_stepwise, which = 1, col = "blue")
```



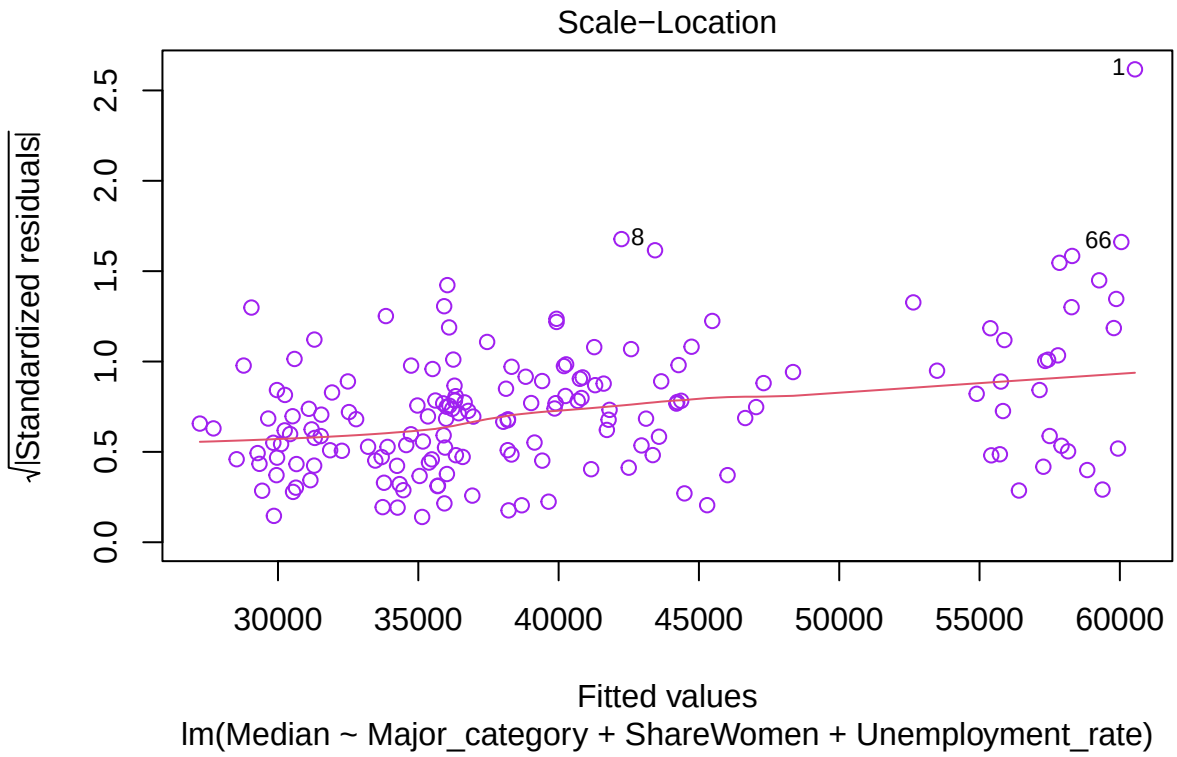

```
plot(model_stepwise, which = 2, col = "green")
```

```
## Warning: not plotting observations with leverage one:
##   109
```

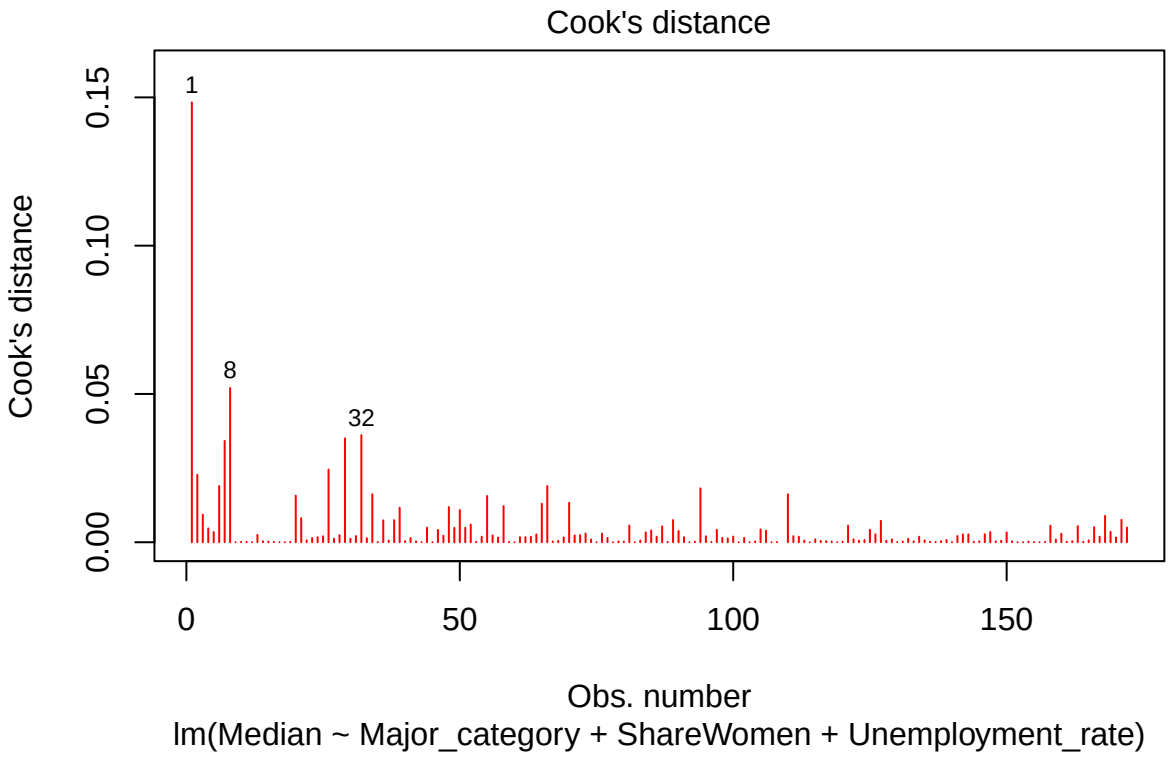



```
plot(model_stepwise, which = 3, col = "purple")
```

```
## Warning: not plotting observations with leverage one:
##   109
```




```r
plot(model_stepwise, which = 4, col = "red")
```




## Model Diagnostics:

Residuals vs. Fitted Plot: The residuals are scattered fairly evenly around zero, which suggests that the linearity assumption is met. There is no strong curvature or obvious pattern. The spread increases slightly at higher fitted values, indicating mild heteroscedasticity, but not enough to raise major concerns about the model.

Normal Q–Q Plot: Most of the points fall close to the reference line, showing that the residuals are approximately normally distributed. There are some variations in the upper tail, probably because of a few high-earning majors, but overall the normality assumption appears acceptable.

Scale–Location Plot: The Scale–Location plot shows a slight upward trend, which means the variance of the residuals increases as the fitted values get larger. This suggests some mild heteroscedasticity, but the pattern isn't strong enough to be a major issue. Overall, the model still is consistently across the range of fitted values

Cook's Distance Plot: A few observations stand out as more influential (such as points 1, 8, and 32). These majors have a larger impact on the model's coefficients, but none of the Cook's Distance values approach the cutoff of 1. This means that while these points are influential, no single observation is overly affecting the model

Overall Assessment: Altogether, the diagnostic plots show that the model assumptions are reasonably satisfied. here is some mild heteroscedasticity and a few influential observations, but nothing severe enough to affect the overall reliability of the model. The regression remains reliable for interpreting how the predictors relate to median salary.

```r
# Identify influential points
cooksd <- cooks.distance(model_stepwise)
which(cooksd > 4/length(cooksd))
```

```
##  1  7  8 26 29 32
##  1  7  8 26 29 32
```

**Influential Observations (Cook's Distance):**

To check for observations that might have an unusually strong impact on the regression, we calculated Cook's Distance for each major and flagged any values above the common cutoff of 4/n. The results show that observations 1, 7, 8, 26, 29, and 32 exceed this threshold. These majors influence the fitted model more than the average observation, either because they have higher leverage or larger residuals.

Influential points aren't necessarily errors, but they can still affect the coefficient estimates and how stable the model is. These results line up with the Cook's Distance plot, which showed noticeable spikes at the same observation numbers. Since none of the Cook's Distance values come close to 1, no single point has an extreme impact on the model, but it's still worth recognizing that these observations play a larger role in shaping the regression results.

```r
#Colinearity
vif(model_stepwise)
```

```
##                       GVIF Df GVIF^(1/(2*Df))
## Major_category    3.746539 15        1.045011
## ShareWomen        2.934076  1        1.712915
## Unemployment_rate 1.297024  1        1.138870
```

**Multicollinearity Check (VIF Analysis):**

To check for multicollinearity in the stepwise model, I calculated the Variance Inflation Factors (VIFs). All of the values were low: Major_category had a GVIF-adjusted value of 1.045, ShareWomen had a VIF of 1.71, and Unemployment_rate had a VIF of 1.14. These are well below the usual cutoffs of 5 or 10, so multicollinearity is not a concern here.

This shows that the predictors don't overlap much in the information they provide, and each one adds something meaningful to explaining differences in median salary

```r
#Nested Model Comparison
anova(simple_model, model_stepwise)
```

```
## Analysis of Variance Table
##
## Model 1: Median ~ ShareWomen
## Model 2: Median ~ Major_category + ShareWomen + Unemployment_rate
##   Res.Df        RSS Df  Sum of Sq      F    Pr(>F)
## 1    170 1.3865e+10
## 2    154 8.4937e+09 16 5371101516 6.0865 2.824e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

$$F = \frac{(\text{RSS}_{\text{reduced}} - \text{RSS}_{\text{full}})/(p_{\text{full}} - p_{\text{reduced}})}{\text{RSS}_{\text{full}}/(n - p_{\text{full}})}$$

**Nested Model Comparison (ANOVA Test):** To compare the simple model with the stepwise model, we ran a nested ANOVA. The simple model includes only ShareWomen, while the stepwise model adds Unemployment_rate and Major_category.

The ANOVA results show a large drop in the residual sum of squares when moving to the stepwise model, and the p-value is extremely small. This tells us that the stepwise model provides a significantly better fit. Adding Major_category and Unemployment_rate explains much more of the variation in median salary than ShareWomen alone.

This confirms that the stepwise model is a meaningful improvement. This also matches the increase in $R^2$ (from 0.3828 to 0.6219), showing that including major category greatly strengthens the model's ability to explain differences in median salary.

## Conclusion:

In this project, we looked at how gender, unemployment rate, and academic field relate to median salary outcomes for recent college graduates. Our simple linear regression showed a strong negative relationship between the proportion of women in a major and median salary, explaining about 38% of the variation in earnings. Once we added major category and unemployment rate in the multiple regression model, the effect of ShareWomen became smaller but stayed statistically significant, meaning gender composition still plays an important role even after accounting for field of study.

Major category ended up being the strongest predictor of salary. Fields like Engineering, Business, Computers & Mathematics, Health, and Law & Public Policy were associated with much higher earnings compared to the baseline category. On the other hand, unemployment rate didn't add much once the other variables were included. The multiple regression model explained over 62% of the variation in median salary, which is a big improvement over the simple model. The stepwise selection process also supported this, showing that major category and ShareWomen were the key predictors, while Total didn't improve the model.

Our plots showed that the model assumptions were mostly met, with only mild heteroscedasticity and a few influential points that didn't exceed standards. Overall, our analysis suggests that academic field is the main driver of salary differences across majors, and gender composition also plays a meaningful role. These results highlight the importance of considering both field of study when looking at how different majors and student characteristics relate to earnings.

## References

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